

[25/06/2026 11:41 PM] hssnd: Dr. Doc:

👤🏠 Welcome to Biochemistry class 🌱

☐ CVS BIOCHEMISTRY ☐

I. Overview of Cardiomyocyte Metabolism

1.1 Definition and Core Principles

☐ Cardiac Metabolism:

▲ This is the integrated biochemical process of producing and utilizing chemical energy (ATP) to power ⚡ the heart's ☐ mechanical work.

▼ It also involves the expression of various transporters 🚚 that regulate the homeostasis of energy ☐ substrates and xenobiotics(foreign chemicals like drugs 🍬)

✓ This metabolic activity is the fundamental prerequisite for the heart's ability to continuously pump oxygenated blood ☐ to the rest of the body.

▲ The heart must maintain a high rate of ATP production to match its continuous utilization. Aerobic production is required for the -ATPase, as well as for the contraction and relaxation cycles of the myocardium.

2. General Features of Cardiac Metabolism

✂ Characteristics of Cardiac Muscle

☐ The heart possesses unique physiological adaptations to support its relentless workload:

🔄 Fatigue Resistance:

👉 Cardiac muscle is specifically adapted to be highly resistant to fatigue, allowing it to function without rest for a lifetime. ➡️ SOON

🔄 Large ⬆️ Myoglobin 🌐 Content:

👉 The heart contains high levels of myoglobin, a single polypeptide heme protein that reversibly binds oxygen.

👉 It serves as an oxygen storage protein, binding and releasing oxygen based on cellular concentration to supply myocytes.[Function]

👉 It is also responsible for the red color 🌐 of cardiac muscle. ☐ [Appearance]

🔄 Continuous + Aerobic Production:

👉 The heart relies almost entirely on aerobic metabolism(99%), except for very short periods of extreme exertion.

👉 Its robustness is evidenced by having the highest oxygen consumption rate per unit weight in the body.

✳️ Cardiac muscle ☐ predominantly relies on aerobic metabolism to generate ATP.

🌀 Limited Anaerobic Metabolism: Ⓞ

👉 The heart has a limited ability for anaerobic metabolism.

 Normal Conditions:

✓ Only about 1% of energy is derived from anaerobic sources.

 Hypoxic Conditions:

✓ This can increase to 10% under moderate hypoxia.

☐ The heart is tuned to aerobic👉 metabolism that it is unable to pump effectively in ischemic condition.

🌀 Massive ☒ ATP Turnover:

👉 The heart exhibits a very high turnover rate; the amount of ATP turned over in a single day is 15 to 20 times the heart's own weight.

🌀 High ☒ ATP Hydrolysis:

👉 Continuous mechanical activity results in many ATP hydrolysis events, which are essential to power membrane transport and contraction-relaxation cycles.

🌀 Low ☐ ATP Storage:

👉 ATP storage is minimal, making the heart dependent on real-time production.

🌀 Minimal ☐ Energy Reserves:

👉 The heart operates on a "just-in-time" basis.

👉 The heart contains negligible energy reserves in the form of glycogen or lipid.

✓ Tight Metabolic Coupling:

▼ Because ATP storage within the cardiomyocyte is minimal, there must be a tight coupling between ATP production and myocardial contraction to maintain normal function.

Large ☒ Mitochondrial Content:

👉 Cardiac myocytes are richly packed with mitochondria, which occupy 1/3 of the cell volume or up to 40-50% of the cytoplasmic space.

👉 The heart contains highest mitochondrial density among all cells in the body.

👉 This high density supports its massive aerobic requirements.

3. Cardiac Energetics

☐ Cardiac Energy Requirement

Daily Demand:

👉 The energy requirements of the heart are exceptionally high, requiring the equivalent of 6 kg of ATP per day in a human heart.

Continuous Contraction:

⊖ Unlike skeletal muscle, the heart must contract continuously without rest for the duration of a lifetime.

ATP Generation Overview

Mitochondrial Dominance:

 The cardiac mitochondrion generates >95% of the ATP used by the heart.

Oxidative Phosphorylation:

▲ ATP is produced continuously via oxidative phosphorylation in the inner mitochondrial membrane.

[25/06/2026 11:41 PM] hssnd: ▪Electrons from NADH and FADH₂ are transferred through the Electron Transport Chain (ETC), pumping protons to create an electrochemical gradient that drives Complex V (ATP synthase) to phosphorylate ADP into ATP.

☒ Substrate Pathways:

 Substrates like fatty acids, glucose, lactate, and ketone bodies are converted into Acetyl CoA, which enters the TCA cycle to generate the electron donors (NADH and FADH₂) required for the ETC.

▼ Mitochondrial Regulation:

✓ Beyond ATP, mitochondria also regulate

 intracellular calcium homeostasis

 cell signaling  via ROS  What the .

 Mitochondria produce small amount of ROS. At low and controlled levels, ROS act as signaling molecules (Messengers) 

 cell death (Apoptosis) if cytochrome C leaks from IMM it induce 

 Functional Energy Demand

 The heart has a massive and unrelenting demand for energy to maintain its physiological functions.

 This demand is categorized by how the generated ATP is utilized:

 Major ATP Consumers:

 Approximately 70% of the total energy generated is dedicated to the mechanical work of contraction & relaxation cycle.

 while the remaining 30% powers ion pump functions, such as the Na^+/K^+ -ATPase and Ca^{2+} uptake by the sarcoplasmic reticulum (SERCA).

 Na^+/K^+ -ATPase:

 ATP is required to power the sodium-potassium pump, which maintains the electrochemical gradients necessary for excitability. 

 Contraction and Relaxation:

 Continuous aerobic production is essential for both the contraction and relaxation cycles of the myocardium.

 Sarcoplasmic Reticulum Uptake (SERCA):

 Specifically, ATP is used for uptake by the sarcoplasmic reticulum via the SERCA pump, which is vital for cardiac relaxation.

 Metabolic Response to Work:

 Because the heart must relax and contract rhythmically, any increase in these activities leads to a direct and corresponding increase in myocardial metabolism to meet the higher ATP demand.

▲ Energy Transfer (The Creatine Kinase System):

◆ The predominant mechanism for transferring energy from the mitochondria to consumption sites is the creatine kinase (CK) system.

 Forward Reaction:

👁️ Mitochondrial CK generates phosphocreatine (PCr) via the phosphorylation of creatine within the mitochondrial intermembrane space.

🕒 Phosphocreatine Properties:

✓ PCr is relatively less polar than ATP and can be stored  at high levels in the cytosol.

 Reverse Reaction:

✓ PCr is available for the rapid renewal of ATP during times of high need; this reverse reaction is catalyzed by cytosolic isoforms of CK at sites of energy consumption, such as sarcomeric proteins and ion channels.

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▼ Deregulation of Metabolism

✓ When cardiac metabolism is deregulated, the heart's ability to function as an efficient pump is compromised.

💀 Pathological Progression:

☐ Various common cardiovascular diseases can lead to deregulation, which ultimately results in heart failure (HF).

 Reduced Efficiency:

✂ Under conditions like ischemia, there is a decrease in oxygen and nutrients.

+ This activates AMPK (due to an increased AMP/ATP ratio), which attempts to
+ stimulate glucose uptake and glycolysis.

 However, this often leads to increased proton production, which results in decreased cardiac efficiency.

👁️ Metabolic Dysfunction: ⌚

✓ The heart is so finely "tuned" to aerobic metabolism that it is unable to pump effectively under ischemic conditions

[25/06/2026 11:44 PM] hssnd: [Photo]

▲ #Normal vs □Heart failure ⊙ metabolism

👤 \$ Dr.Doc □

[25/06/2026 11:44 PM] hssnd: □Biomarkers of Myocardial □ Injury. ⊙

▲ Cardiac biomarkers are protein molecules released into  the bloodstream □ from damaged heart muscle following a characteristic rise and fall pattern.

📄 These markers are natural intracellular  components in myocytes, they are not supposed to be found in the blood like K^+/Na^+ , So if a □ blood test  indicate their presence **+** this suggest the death  of some myocytes (myocardial infarction)

📌 Myocardial Infarction Pathway

~\ MI-Pathway

🔑 Myocardial infarction (MI) follows a specific pathological progression:

1. Coronary Occlusion: ⊙

⊖ Occlusion of coronary artery. (Blockage of the coronary arteries.)

2. Ischemia:

— This occlusion leads to  a restricted blood □ and oxygen  supply to heart tissue. □

3. Infarction:

 Prolonged ischemia causes irreversible damage to the heart tissue.

4. Release of Biomarkers  □:

▪ As the cell membranes of the damaged **X** myocytes fail, intracellular enzymes and proteins leak into the blood. □

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#Dx □ of MI □

📄 Criteria For Diagnosis 🔍 Of MI □ ⊙

🌀 Requires the presence of At least 2 of the following characteristics :

1. Typical heart □ attack ⊙ symptoms. 

2 Characteristic Pattern of a cardiac marker   in the plasma □

☒ Rise and gradual fall  of cardiac troponins

 More rapid rise and fall of CK-MB

3. Typical ECG pattern

▲ e.g. elevated ST segment

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#Characteristics/ Features of ideal cardiac marker.

☐ Features of an ideal cardiac marker.

☐ High cardiac specificity

✓ Easy diagnosis

✓ Marker should play a designed role in the treatment ☐ and management of clinical subject.

[25/06/2026 11:46 PM] hssnd:  Classification of cardiac biomarkers ☐

1. Biomarkers of myocardial injury

▲ Cardiac troponins

 cTnI

 cTnT

▲ Creatine kinase-MB

▲ Myoglobin

2. Biomarkers of hemodynamic stress

▲ Natriuretic peptides

 ANP

 B-NP

 C-NP

3. Inflammatory and prognostic biomarkers

☒ C-reactive protein

☒ Myeloperoxidase

☒ Homocysteine

4. Heart tissue Ischemia Marker.

▲ h-FABP

▲ IMA

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▲ Creatine Kinase (CK)

📖 Structure and Isoenzymes:

✧ CK is an enzyme expressed in various tissues, consisting of two subunits(Dimer).

✓ No. Of Subunits: There are 2 subunits. CK have 4[°] structure.

□ Definition of quaternary structure.

✧ This level of organization only present if a protein is made up of 2/more polypeptide chain(Subunits) working together as one unit.

✓ Types of subunits: There are 2 types

□ B (brain type) and

🦶 M (muscle type).

✧ These combine to form three different isozymes,are similar in function but d/t in structure(sub-Unit):

~\CK-MM:(CK-3)

🦶 Predominates in skeletal muscle. 🦶

~\CK-BB:(CK-1)

🦶 Predominates in the brain, smooth muscle, and lungs.

[found mostly in the Brain□]

~\CK-MB: (CK-2)

🦶 Predominates in cardiac muscle and has high specificity for heart tissue.[found mostly in the heart□]

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 Function:

 It catalyzes the conversion of creatine to phosphocreatine.

 CK-MB Timeline:

 Rise:

✓ Begins 4–6 hours after the onset of infarction.

 Peak:

✓ Approximately 12 hours.

 Normal:

✓ Returns to baseline within 24–36 hours.

▲ Advantages

✓ Useful for early diagnosis of MI.

✓ Useful for diagnosis of re-infarction

✓ Reinfarction Marker:

🕒 Because CK-MB returns to baseline relatively quickly (within 48 hours), it is the preferred marker to indicate early re-infarction if levels normalize and then increase again.

▲ Just to be clear 🔍

CK-MB is Useful for diagnosis 🔍 of re-infarction. After 3 days of MI CK-MB would disappear while troponin will be low. So if we detect elevated CK-MB and low troponin it means a second episode of MI.

▼ Disadvantages

✓ Not significant if measured after 2 days of MI (delayed admission)

✓ Specific but Not highly specific (elevated in skeletal muscle damage)

 Relative index ☐

CK-MB mass $\times 100$ / Total CK

® Relative index More than 5 % is indicative  for MI.

⚠ Extreme exercise, especially in unfit individuals, causes an elevated CK, potentially to very high levels.

[25/06/2026 11:47 PM] hssnd: [Photo]

[25/06/2026 11:47 PM] hssnd: ▲ Troponins

▫ Troponin is a complex of three regulatory proteins integral to muscle contraction, regulating the interaction between  actin and myosin.

▣ Subunits:

▪ TnC (Calcium Binding):

▼ Binds calcium to initiate contraction but has no ✕ diagnostic value due to homology between cardiac and skeletal isoforms.

⊖ TnI (Inhibitory):

© Unique to the heart ☐ and the most cardiac-specific.

✓ Highly specific markers for detecting MI.

☐ It contains additional amino acid residues on its N-terminal not found in skeletal forms.

▲ Has 3 isoforms (1 cardiac + 2 skeletal)

 TnT (Structural):

✚ Attaches the troponin complex to tropomyosin and actin.

🔍 Cardiac TnT (cTnT) is present + in fetal skeletal muscle but absent in healthy adult skeletal muscle, but reexpressed during HF in damaged skeletal tissue.

 Timeline:

 Rise:

🕒 Begins 2–3 hours after injury.

▲ Peak:

 12–24 hours.

— Persist:

~~ Elevated levels can remain in the blood for 5–10 days

☐ Has similar t_{1/2}(2-4hrs) & early serum increase  (2-8hrs) as that of cTnI

💬 Multiple isoforms in skeletal muscle tissue & 4 isoforms in cardiac muscle tissue.(1 adult + 3 fetal tissue)
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💬 #Cardiac ☐ Biomarkers that display biphasic or double-peak release pattern in the bloodstream ☐

☐ In cardiology, Cardiac Troponin T (cTnT) (especially measured by high-sensitivity assays) and Myoglobin are well-documented for displaying a biphasic or double-peak release pattern in the bloodstream following myocardial injury and successful reperfusion.

✳️ While CK-MB also occasionally shows a secondary rise, this is generally a sign of re-infarction or ongoing infarct extension rather than a natural biphasic elimination curve.
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▲ #h-FABP(15KDa) is smaller than any other markers, allowing for its rapid renal clearance from the blood.

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📦 Natriuretic Peptides (NPs)

▲ NPs are structurally ☐ similar but genetically distinct ✕ neurohormones that maintain cardiovascular homeostasis.

⚙️ They act as markers ☒ of cardiac stress. [Functional Markers]

▲ ANP (Atrial Natriuretic Peptide):

✓ Synthesized in the atria ▲ and released in response to atrial distension 🌀.

▲ BNP (B-type Natriuretic Peptide):

✓ Mainly produced by ventricular myocytes following volume overload and ventricular wall stretch.

☐ In healthy states, circulating levels are low, becoming notable only in pathology.

▲ CNP (C-type Natriuretic Peptide):

📦 Found in the brain and synthesized by vascular endothelial cells.

⚡ Triggered by cytokines like IL-1 and TNF.

☐ Likely outcome from release of natriuretic peptides [Functions]:

⊖ Antagonism:

⊖ They suppress the Renin-Angiotensin-Aldosterone System (RAAS) and Sympathetic Nervous System (SNS)

↔ Vasodilation:

☒ Decreasing peripheral vascular resistance to lower blood pressure.

☒ ↑ TOP Increased Renal Blood Flow & GFR.

☒ Natriuresis and Diuresis:

☒ Promotion of sodium and water loss.

☒ Myocardial Relaxation:

☒ → SOON Direct promotion of relaxation.

☒ Inflammatory Cardiac ☐ Biomarkers

✓ C-Reactive Protein (CRP)

☆ Structure:

☒ An acute-phase protein produced by the liver with a pentameric structure (five identical 23 kDa subunits).

▼ Functions:

✓ Useful marker for cardiovascular risk because atherosclerosis is characterized by chronic arterial inflammation.

☐ Mechanism of Atherogenesis:

⚡ Activates the classical complement pathway.

✚ Stimulates phagocytosis.

▲ Causes endothelial dysfunction by increasing nitric oxide (NO) synthesis.

✚ Increases LDL deposition in plaques via stimulated macrophages

▪ Less Common Cardiac Markers

✓ Lactate Dehydrogenase (LDH):

▲ LDH Reaction:

👉 Catalyzes the reversible conversion of pyruvate to lactate.

▲ Isoenzymes:

🌀 Exists in five forms.

□ LDH-1 is found primarily in heart muscle.

□ LDH

💬 No. Of sub units: 4(Tetramer). So have quaternary structure.

💬 Type of sub units: 2

#H(heart)

#M(Muscle)

👉 The 5 Combinations

▼ LDH-1(HHHH)

® 4 heart sub — units

® Found in heart & RBCs □

© Dominant in the heart □

▼ LDH-2(HHHM)

© 3 heart + 1 muscle subunit

® Found in RES

▼ LDH-3(HHMM)

© 2 heart + 2 muscle subunit

© Dominant in the Lung □

▼ LDH-4(HMMM)

© 1 heart + 3 muscle subunit

© Found in kidney + pancreas

▼ LDH-5(MMMM)

© 4 muscle subunits

© Liver + skeletal muscle

👉 Historical Role:

✓ Widely used in the past; an increase in LDH-1 compared to LDH-2 indicates an AMI.

□ In healthy individuals $[LDH-2] > [LDH-1]$ in the blood□.

📄 During MI heart release massive amount of LDH-1
 $[LDH-1] > [LDH-2]$ in the blood□.

□ They are detectable in serum 12–24 hours after a heart attack.

🌀 Myeloperoxidase:

👉 An inflammatory marker.

🌀 Homocysteine:

👉 Associated with cardiovascular risk and severity of disease.

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👤 Lipid digestion and transport(part-1)

🌀 Digestion is the process of converting complex and hydrophobic dietary lipids into absorbable units & into a form that can travel through the aqueous environment of the bloodstream .

🕒 Represent the essential "entry phase" where hydrophobic dietary fats are converted into a form that can travel through the aqueous environment of the bloodstream to reach various tissues.

🕒 Lipid Digestion and Absorption

📖 The digestion of lipids involves both mechanical and chemical processes to break down complex triacylglycerols into absorbable units.

👤 🏠 Hormonal Trigger:

□ The digestion of lipid is carried out by hydrolytic enzyme and the secretion of these lipolytic enzymes is regulated by hormone. The process is largely regulated by the hormone Cholecystokinin (CCK).

□ Cholecystokinin release

👉 stimulates the pancreas to secrete digestive enzymes(pancreatic juice) and triggers the gallbladder to contract, releasing bile into the duodenum.

✓ Chemical Breakdown:

 Pancreatic lipase is the primary enzyme responsible for the chemical breakdown of fats.

 It hydrolyzes triacylglycerols into 2 free fatty acids and 2-monoacylglycerol, which serves as an obligatory intermediate for fat absorption.

 **Micelle Formation:**

 Because long-chain fatty acids are insoluble in water, they require a specialized transport vehicle to move from the intestinal lumen to the cell surface.

 This is achieved through the formation of a bile salt micelle, a molecular complex that carries fatty acid residues to the surface of the gut epithelial cells (enterocytes) for absorption.

 **Clinical Correlation:**

 A deficiency in these processes—such as a lack of Cholecystokinin release or pancreatic enzymes—leads to fat malabsorption, resulting in steatorrhea (excess fat in the feces) and related issues like night-blindness due to poor absorption of fat-soluble vitamins like VA.

 **Initial Transport:**

 **Synthesis and Maturation of Chylomicrons**

 Once lipids are absorbed by the enterocytes, they must be packaged for transport through the lymphatic system and the blood.

 **Chylomicron Synthesis:**

 Inside the intestinal cells, the absorbed fatty acids and 2-monoacylglycerol are re-synthesized into triacylglycerols and packaged into large lipoprotein particles known as nascent chylomicrons.

 **The Journey:**

 These nascent chylomicrons are released from the intestinal cells into the lymphatic vessels and eventually enter the bloodstream.

 **Maturation Process:**

 In the blood, nascent chylomicrons are not yet ready for tissue utilization.

 They must be converted into mature chylomicrons.

 **Role of HDL:**

 This maturation occurs when the chylomicrons acquire specific proteins—Apo protein E and C-II—from HDL (High-Density Lipoprotein).

□ Tissue Activation:

📖 The acquisition of ApoC-II is critical because it acts as a cofactor that activates the enzyme

📖 Lipoprotein Lipase (LPL).

📖 LPL is located on the surface of peripheral tissues (like adipose and muscle) and allows them to break down the chylomicron's triacylglycerols for energy or storage.

#Lipidabsorptionandtransport.

✓ Lipoproteins

📖 are specialized Macromolecular complexes that serve as the primary transport vehicles for hydrophobic lipids through the aqueous environment of the bloodstream.

📖 coordinated metabolism of chylomicrons and other major lipoproteins. 📖

📖 Overview of Lipoprotein Classification

📖 Lipoproteins are classified based on their density, composition, and origin.

✓ Chylomicrons:

📖 Transport exogenous (dietary) lipids from the intestine to peripheral tissues.

✓ VLDL (Very-Low-Density Lipoprotein):

📖 Produced by the liver to transport endogenous triacylglycerols.

✓ LDL (Low-Density Lipoprotein):

📖 Primarily transports cholesterol from liver to peripheral tissues; known as "Bad Cholesterol".

✓ HDL (High-Density Lipoprotein):

📖 Facilitates reverse cholesterol transport (moving cholesterol from tissues back to the liver); known as "Good Cholesterol".

[25/06/2026 11:49 PM] hssnd: 📖 Chylomicron Metabolism:

📖 📖 The Exogenous Pathway

☆ This pathway describes how dietary fats reach the body's cells.

📖 Micelle Formation & Absorption:

📖 Long-chain fatty acids are carried from the gut lumen to the surface of enterocytes within bile salt micelles for absorption.

🗂️ Nascent Chylomicron Synthesis:

📖 Once inside the intestinal cells, lipids are re-packaged into nascent chylomicrons.

📖 These particles contain Apo-B48, which is unique to lipoproteins of intestinal origin.

🗂️ Maturation via HDL:

📖 To become functional, nascent chylomicrons must be converted into mature chylomicrons.

📖 This maturation requires the acquisition of Apoprotein E and Apo C-II, both of which are donated by HDL in the circulation. So HDL is called Repository of Apoprotein E and Apo C-II.

🗂️ LPL Activation:

🕒 ApoC-II is a vital cofactor that activates the enzyme Lipoprotein Lipase (LPL) located on the capillary walls of adipose and muscle tissues.

🕒 This activation allows the release of fatty acids for tissue utilization.

🗂️ Chylomicron Remnants:

📖 After LPL removes most of the triacylglycerols, the remaining particle is a chylomicron remnant, which is cleared by the liver through ApoE recognition.

🗂️ VLDL and LDL:

👤🏠 The Endogenous Pathway

🔑 This pathway manages lipids synthesized within the liver.

📁 VLDL Production:

📖³¹ The liver synthesizes VLDL to transport triacylglycerols, often produced from excess dietary carbohydrates; hence, a low-carbohydrate diet is beneficial for patients with elevated VLDL.

📁 Apoprotein Content:

🗂️ Unlike chylomicrons, VLDL and its derivatives (IDL & LDL) contain Apo-B100.

📄 Conversion to LDL:

 As VLDL loses triacylglycerols via LPL activity, it eventually becomes LDL.

 Cellular Uptake:

 LDL is removed from circulation by the LDL receptor, which is specific for both Apo B-100 and Apo E.

 HDL and Reverse

 Cholesterol Transport

 HDL plays a protective role by managing the body's cholesterol levels.

 *Protein Reservoir:

 HDL acts as a circulating reservoir for Apo C-II and Apo E, which are essential for the maturation of both chylomicrons and VLDL.

 Reverse Transport:

 HDL picks up excess cholesterol from peripheral tissues and returns it to the liver.

 This process is enhanced by the ABCA1 protein, which moves cholesterol out of cells so HDL can capture it.

 Clinical Correlations and Disorders

 Defects in lipoprotein metabolism lead to significant medical conditions:

 Abetalipoproteinemia:

 A genetic inability to produce chylomicrons, which results in a failure to absorb fat-soluble vitamins (like Vitamin K), leading to symptoms such as difficulty in blood clotting.

 Type III Hyperlipidemia:

 Caused by a deficiency of ApoE, resulting in the dangerous accumulation of elevated chylomicron and VLDL remnants in the blood.

 Steatorrhea:

 Caused by deficiencies in fat digestion/absorption (e.g., inhibition of Cholecystokinin release), leading to excess fat in stools and fat-soluble vitamin deficiencies.

#Lipoproteins

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[25/06/2026 11:50 PM] hssnd: In the gastrointestinal (GI) tract, the lymphatic system plays a

critical role in the absorption and transport of dietary fats, particularly long-chain fatty acids, which are too large to directly enter blood capillaries.

✓ Key Components and Processes:

🔑 Lacteals: These are specialized lymphatic capillaries located in the center of each intestinal villus in the small intestine. They are responsible for absorbing dietary lipids.

🔑 Chyle: After a meal, the lymph within these vessels becomes a milky-white fluid called chyle, which is rich in emulsified fats.

🔑 Chylomicrons: Inside the intestinal cells (enterocytes), fatty acids and monoglycerides are re-esterified into triglycerides and packaged with cholesterol and proteins to form large lipoprotein particles called chylomicrons.

📦 Transport Mechanism:

Because of their large size, chylomicrons are released via exocytosis into the interstitial fluid and enter the lacteals through junctions. They are then transported via the mesenteric lymphatic ducts and the thoracic duct, eventually entering the bloodstream through the subclavian vein.

🔑 Role of Smooth Muscle: The contraction of smooth muscles surrounding the lacteals helps pump the fat-rich lymph.

✓ This pathway allows fat to bypass the liver and enter the general circulation directly.

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☐ Lipoprotein metabolism.

✳️ serves as the master link coordinating the transport of hydrophobic lipids—triacylglycerols and cholesterol—through the aqueous environment of the bloodstream. This process is governed by specific "players," enzymatic gatekeepers, and apoprotein "keys" that dictate tissue delivery and clearance.

👤 The Players: Lipoprotein Classification

📖 Lipoproteins are spherical complexes classified by their density and origin:

✳️ VLDL (Very-Low-Density Lipoprotein):

🌀 Synthesized in the liver to transport endogenous triacylglycerols (TAGs) to peripheral tissues.

📖 Mature VLDL contains

☢️ TAGs,

☢️ cholesterol,

- ☢ Apoproteins B-100,
- ☢ Apoproteins C-II, and
- ☢ Apoproteins E, but notably ☹ lacks Apo-B48.

✳ IDL (Intermediate-Density Lipoprotein):

☹ The transient remnant of VLDL after ☹ TAGs are removed.

✳ LDL (Low-Density Lipoprotein):

○ Derived from IDL, it is the primary carrier of cholesterol to the tissues. Known as "Bad Cholesterol," high levels are associated with atherosclerosis.

⚡ Small dense LDLs are particularly atherogenic because they remain in circulation longer and are prone to oxidation.

✳ HDL (High-Density Lipoprotein):

⬤ Known as "Good Cholesterol," it functions in reverse cholesterol transport, moving excess cholesterol from peripheral tissues back to the liver for excretion.

⬤ It also serves as a circulating reservoir for Apo C-II and Apo E.

☞ The Enzymatic Gatekeeper.

☢ Two key enzymes control the modification and utilization of lipoproteins:

👤 Lipoprotein Lipase (LPL):

📍 Located on the capillary walls of adipose and muscle tissues.

🔑 It is responsible for hydrolyzing the TAGs within chylomicrons and VLDL, allowing tissues to use fatty acids for energy or storage.

👤 LCAT (Lecithin-Cholesterol Acyltransferase):

📖 Associated with HDL, this enzyme esterifies cholesterol, allowing HDL to pack and transport it efficiently; a defect in LCAT contributes to HDL depletion.

🔑 The Apoproteins: "Keys" to the Doors

📖 Apoproteins are the protein components of lipoproteins that provide structural integrity and act as ligands for cell receptors:

👤 Apo B-48:

NEW The unique "signature" of chylomicrons, essential for their synthesis and secretion from

the intestine.

 Apo B-100:

 The "signature" of VLDL and LDL, synthesized in the liver. It acts as a ligand for the LDL receptor.

 Apo C-II:

 The essential cofactor activator for Lipoprotein Lipase (LPL).

 It is donated by HDL to nascent chylomicrons and VLDL to trigger fat delivery.

 Apo E:

 A critical ligand required for the liver to recognize and clear chylomicron and VLDL remnants.

 It is also a ligand for the LDL receptor.

☐ Coordinated Metabolism Pathways

 Exogenous Pathway:

 Dietary lipids are packaged into nascent chylomicrons (containing Apo B-48).

 They enter the ☐ blood, acquire Apo C-II and Apo E from HDL, and become mature chylomicrons.

☐ LPL breaks down their TAGs, and the remaining remnants are cleared by the liver via Apo E.

 Endogenous Pathway:

 The liver produces VLDL (containing Apo B-100) from excess carbohydrates.

 VLDL is processed by LPL into IDL and finally LDL. LDL is cleared by the LDL receptor, which recognizes Apo B-100 and Apo E.

 Reverse Cholesterol Transport:

▫ HDL picks up cholesterol from peripheral cells—a process facilitated by the ABCA1 protein—and returns it to the liver.

☐ Clinical Correlation:

☐ Hyperlipidemias and Disorders ☐ 

🧐 Defects in these keys and enzymes lead to specific metabolic diseases:

🧐 🧠 Type III Hyperlipidemia:

🧐 Caused by a deficiency of Apo E, leading to the accumulation of chylomicron and VLDL remnants in the blood.

🧐 🧠 Abetalipoproteinemia:

🧐 A failure to produce lipoproteins containing Apo B (chylomicrons and VLDL), resulting in fat malabsorption and difficulty in blood clotting due to Vitamin K deficiency.

🧐 Metabolic Syndrome:

🧐 ➡ Characterized by

- 1 Elevated Apo B-100,
- 2 hypertriglyceridemia, and
- 3 low HDL.

🧐 Hypertriglyceridemia:

[25/06/2026 11:50 PM] hssnd: 🧐 Often leads to the production of dangerous small dense LDLs and the depletion of HDL.

The End of Lipid Metabolism!

#Lipoprotein

[25/06/2026 11:51 PM] hssnd: [Photo]

▲ #FREDRICKSON LEVY LEES CLASSIFICATION OF HYPERLIPOPROTEINEMIA

[25/06/2026 11:51 PM] hssnd: Dr. Doc:

□ ✨ DYSLIPIDEMIA ✨ □

✨ Disorders of Lipid & Lipoprotein Metabolism ✨

📖 Definition

□ Dyslipidemia refers to abnormalities in:

⚙️ Lipid metabolism

🚚 Lipid transport

□ Lipoprotein synthesis/degradation

➡ Resulting in:

☑ Increased blood lipids

OR

 Reduced blood lipids

CLASSIFICATION

PRIMARY (FAMILIAL)

☐ Due to genetic defects

SECONDARY

Occurs due to other diseases:

 Diabetes Mellitus

 Hypothyroidism

 Obesity

 Alcoholism

☐ I. FAMILIAL HYPERLIPIDEMIAS

 Fredrickson Classification

TYPE I FAMILIAL HYPERCHYLOMICRONEMIA

 Metabolic Defect

✗ Deficiency of:

☐ Lipoprotein Lipase (LPL)

OR

☐ Apo C-II

(LPL activator)

☒ Increased Lipoprotein

 CHYLOMICRONS

☐ Increased Plasma Lipid

☒ TRIACYLGLYCEROL (TAG)

☐ Clinical Features

☐ Creamy plasma appearance

 Atherosclerotic Risk May increase

 Treatment

 Strict low-fat diet

☐ TYPE IIa FAMILIAL HYPERCHOLESTEROLEMIA

 Metabolic Defect

☐ Autosomal dominant mutation causing:

☒ Defective LDL receptors

☒ Increased Lipoprotein

 LDL

☐ Increased Plasma Lipid

☒ CHOLESTEROL

 TAG usually normal

 Atherosclerotic Risk

 VERY HIGH 

Especially:  Coronary arteries ☐ Accelerates atherogenesis

 Treatment

 Low-cholesterol diet

 Cholestyramine

TYPE IIb FAMILIAL COMBINED HYPERLIPIDEMIA

 Metabolic Defect

☒ Overproduction of ApoB

+

☒ Decreased LDL receptor activity

☒ Increased Lipoproteins

 LDL

 VLDL

☐ Increased Plasma Lipids

☒ Cholesterol

☒ TAG

 Atherosclerotic Risk

 VERY HIGH

☐ TYPE III FAMILIAL DYSBETALIPOPROTEINEMIA

 “Broad Beta Disease”

 Metabolic Defect

 Defective Apo E synthesis (particularly Apo E2)

☒ Increased Lipoprotein

 IDL

(Intermediate Density Lipoprotein)

☐ Increased Plasma Lipids

☒ Cholesterol

☒ TAG

☐ Classic Feature

 Broad β -band on electrophoresis

 Atherosclerotic Risk

 Very high especially: ☐ Peripheral vessels

 Treatment

 Low-fat diet

 Low-calorie diet

 Clofibrate

 TYPE IV FAMILIAL HYPERTRIACYLGLYCEROLEMIA

 Metabolic Defect

☒ Overproduction of endogenous TAG by liver

☐ Increased VLDL production

☐ Reduced elimination

☒ Increased Lipoprotein

 VLDL

☐ Increased Plasma Lipid

☒ TAG

 Atherosclerotic Risk

 May or may not increase

 Treatment

 Low-fat diet

 Low-calorie diet

 Niacin

TYPE V FAMILIAL COMBINED HYPERLIPIDEMIA

 Metabolic Defect

☒ Increased VLDL production

+

☐ Decreased LPL activity

☒ Increased Lipoproteins

 VLDL

 Chylomicrons

☐ Increased Plasma Lipid

 TAG

 Associations

 Obesity

 Diabetes mellitus

 Atherosclerotic Risk

 VERY HIGH

II. FAMILIAL HYPOPROTEINEMIAS

ABETALIPOPROTEINEMIA

 Cause

 Defect in ApoB synthesis

☐ Autosomal recessive

 Absent Lipoproteins

 LDL

 VLDL

 Chylomicrons

 Consequences

☐ TAG accumulation in: Liver

 Intestine

☐ Clinical Features

 Low cholesterol

 Failure to thrive

☐ Mental retardation

FAMILIAL HYPOBETALIPOPROTEINEMIA

 Cause &  Clinical Importance

☒ An inherited impairment in ApoB synthesis where LDL levels are 10–50% of normal. This condition is generally harmless

☐ TANGIER DISEASE

✱ Familial Alpha-Lipoprotein Deficiency

⚙ Key Defect

✕ ABSENCE OF HDL

⚠ Consequence

✕ Impaired reverse cholesterol transport

☐ Result

☒ Cholesteryl ester accumulation in tissues

⚠ Risk

⚠ Increased atherosclerosis

☐ III. OTHER DYSLIPIDEMIAS & MARKERS

☐ LIPOPROTEIN(a) EXCESS

✱ Lp(a)

⚙ Mechanism

[25/06/2026 11:51 PM] hssnd: Contains: ☐ Apo(a) Which resembles: ☐ Plasminogen

⚠ Effect

⊖ Inhibits clot dissolution

☒ Promotes thrombosis

⚠ Clinical Importance

Major contributor to:

♥ CORONARY HEART DISEASE (CHD)

☐ LCAT DEFICIENCY

⚠ Result

☒ Increased ratio of:

- ☐ Free cholesterol
- to
- ☐ Cholesterol esters

☐ HEPATIC LIPASE DEFICIENCY

 Result

☒ Elevated chylomicrons

☐ CETP DEFICIENCY

 Effect

☒ Decreased reverse cholesterol transport

IV. PHYSICAL SIGNS OF HYPERLIPIDEMIA

☐ XANTHOMATA

☐ Lipid deposits in:

 Skin

☐ Tendons

XANTHELASMA PALPEBRARUM

☐ Yellow lipid plaques around eyes

CORNEAL ARCUS

☐ White ring around cornea

LIPEMIA RETINALIS

 Milky retinal vessels

V. TREATMENT PRINCIPLES

THERAPEUTIC LIFESTYLE CHANGES (TLC)

Core Measures

 Weight reduction

 Smoking cessation

 Exercise Recommendation

At least: of moderate-intensity exercise most days

 DIETARY MODIFICATION

Heart-Healthy Diet

☒ Saturated fat: <7% of calories

☒ Low trans fat intake

☒ High fiber intake

☒ Omega-3 fatty acids 

 PHARMACOLOGIC THERAPY

☐ STATINS

HMG-CoA Reductase Inhibitors

 Main Action:

 Lower LDL

Example

 Atorvastatin

☐ FIBRATES

 Main Action:

 Lower TAG

Example

 Fenofibrate

☐ BILE ACID SEQUESTRANTS

Example

 Cholestyramine

☐ NIACIN

 Effects:

 Raises HDL

 Lowers TAG

□ HIGH-YIELD PEARLS ✨

✨ Type I → LPL deficiency → ↑ Chylomicrons 🧪

✨ Type IIa → LDL receptor defect → ↑ LDL 🚚

✨ Type III → ApoE defect → Broad β band 📊

✨ Tangier disease → Absent HDL □

✨ Lp(a) inhibits fibrinolysis □

✨ Statins are first-line for high LDL 💊

✨ Fibrates mainly lower TAG □

✨ Xanthomas = lipid deposits □
